

Project 3: Discontinuous Galerkin

AE 623, Computational Fluid Dynamics II, Winter 2026

Due: March 20, 11:59pm, electronically via Canvas

1 Problem Description

1.1 Introduction

In this project you will extend your finite-volume solver to high-order approximation using the discontinuous Galerkin method. The geometry for this case is the same as in the previous two projects, the turbine blade passage. You should use the sequence of meshes that you generated in the first project, modified to include curved elements.

1.2 Governing Equations

The governing equations are the same as in the finite-volume project: compressible Euler with ratio of specific heats $\gamma = 1.4$. Use the conservative state vector $\mathbf{u} = [\rho, \rho u, \rho v, \rho E]^T$ and convenient units where $\rho_0 = 1$ and $a_0 = 1$ at the inflow. The blade surfaces are slip walls, stagnation quantities are specified at the inflow, and static pressure at the outflow. Use the Roe, HLLE, or similar-accuracy flux (not Rusanov).

1.3 Curved Meshes

For $p > 0$, DG requires curved-element representations of curved boundaries. Only the blade boundaries are curved, so you may find it convenient/efficient to split your elements into two groups: curved ones adjacent to the blade and linear ones everywhere else. The curved elements should be at least quadratic, $q = 2$. Note that you should increase the quadrature order when integrating on curved elements and their boundaries. Also note that the normal vector on curved boundaries is no longer constant along an element's edge.

1.4 Steady-State Simulations

For steady-state simulations, as in the previous project, march to steady state on a given mesh and monitor the L_1 norm of the discrete residual vector. Terminate your calculation when this norm drops five orders of magnitude (10^{-5}) below a baseline residual norm, e.g. that of freestream initialization. For uniform-refinement studies, using the solution from a coarser mesh to initialize the state on a finer mesh will improve robustness. Likewise, for higher-order solutions, you can use a coarser-order solution as the initial guess.

1.5 Unsteady Simulations

In addition to steady-state, consider the same unsteady conditions as in the previous project: the motion of the blades relative to an upstream stator creates an unsteady inflow condition,

$$\begin{aligned}\rho_0(\eta(t)) &= \rho_0 [1 - f_{\text{wake}} \exp(-\eta^2/(2\delta^2))] \\ \eta(t) &= \frac{y_{\text{stator}}}{\Delta y} - \left\lfloor \frac{y_{\text{stator}}}{\Delta y} \right\rfloor - \frac{1}{2} \\ y_{\text{stator}} &= y_{\text{rot}} + V_{\text{rot}} t\end{aligned}$$

where $f_{\text{wake}} = 0.1$, $\delta = 0.1$, $\Delta y = 18\text{mm}$, y_{rot} is the y position on the rotor (i.e. the y position in your domain), V_{rot} is the rotor speed, and $\lfloor \cdot \rfloor$ is the greatest-integer (floor) function. Set the rotor speed to the stagnation speed of sound at the inflow, $V_{\text{rot}} = a_{0,\text{inflow}}$. Start the unsteady simulations with a steady-state solution and run sufficiently long until you observe periodic behavior of the outputs.

1.6 Time Stepping

Use SSP-RK3 or RK4 for time integration. In the steady-state cases, use local time stepping to accelerate convergence. For both steady and unsteady simulations, set the time step using the CFL condition.

1.7 Outputs and Visualization

The outputs remain the same as in the previous project: residual convergence plots, Mach and entropy field contours, c_p distributions, and x and y force coefficients on the blade. You should use quadrature to compute the integrals required for the force calculations. When visualizing a high-order solution, e.g. for the Mach contours, you should show the variation of the solution within each element. You can do this by making a contour plot on a refined mesh, where the states at new points inside elements are computed from interpolating the high-order state.

2 Givens

Several potentially useful files/codes/snippets are provided as part of this project:

- `naca.gri`: An example of a curved mesh of an airfoil, using $q = 3$ curved elements.
- `plotgri.m`: Plots a curved mesh. Note that the `.gri` format allows for multiple element groups and curved elements.
- `matinv.c`: PLU factorization and inversion of matrices.
- `shape.c`: Full-order Lagrange basis functions and gradients for $p = 0, 1, 2, 3$.
- `quad1d.c`: One-dimensional quadrature rules, the Gauss-Legendre points and weights.
- `quad2d.c`: Two-dimensional quadrature rules for a unit right triangle with the right angle point at the origin.

3 Tasks

Address the following tasks in your report.

1. [15%] Write a code that curves linear meshes to $q = 2$ (minimum) or $q = 3$ (better). In your report, include zoomed-in plots of selected $q = 1$ and $q > 1$ meshes from your uniform-refinement sequence. You can use the `plotgri.m` code to make these plots.
2. [30%] Write a DG solver with support for at least $p = 0, 1, 2, 3$. On your coarse mesh and for $p = 0, 1, 2, 3$ (all in turn), perform a free-stream test, a free-stream preservation test, and a run to convergence with the actual boundary conditions – plot the residual norm history versus time step iteration for the actual runs. Describe your implementation and present the results of the tests in your report.
3. [15%] Implement post-processing, including: calculating the outputs; plotting the pressure coefficient distribution; and plotting the Mach number contours. Present these quantities for the coarsest mesh using $p = 0, 1, 2, 3$.
4. [10%] Perform a convergence study using your uniform-refinement mesh sequence (at least the first three meshes, all curved) and tabulate/plot the lift and drag coefficients. Report results starting at $p = 0$ up to the highest order you can afford. Discuss the effect of p on the outputs.
5. [20%] Unsteady runs. Run simulations at orders $p = 0, 1, 2, 3$ on at least two meshes: coarse and fine, for a sufficiently long time horizon to observe several cycles of periodic output

behavior. Show a few entropy field plots at various times and time histories of the force coefficients (overlay some of these to make comparisons). Discuss the effect of mesh resolution and order on the unsteady results.

4 Deliverables

Each group should turn in, electronically via Canvas,

1. A technical report, **12 pages maximum** for this project, as a **.pdf** file that describes your methods and results, and that answers the questions. The report should be professional, complete, and concise. **10%** of your grade will be determined by the presentation: neatness, labeling of axes, spelling, etc.
2. Documented source files of all codes you wrote for this project, as one **.zip** archive. You should provide all files necessary to run your program(s).

This is a group assignment. At the end of the report, summarize the contributions of each group member to the method development, code, results, and report.